

CHOI, Long Ho Happy

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EDUCATION

The Hong Kong University of Science and Technology (HKUST) *Sept 2023 – Present*
Bachelor's Degree in Engineering, major in Computer Science (with an Extended Major in Artificial Intelligence)

- GPA: 3.112/4.3
- Minor in Mathematics
- Minor in Big Data Technology

Tin Ka Ping Secondary School (TKPSS) *Sept 2017 – June 2023*

- 2023 HKDSE:
 - 5* in Mathematics (Compulsory Part), 5* in Mathematics (Extended Part Module 2), 5 in Physics, 5 in Chemistry, 5 in Liberal Studies

PROJECTS & COMPETITIONS

TrustShop: Agent-Orchestrated Online Shopping Assistant (Final Year Project) *Jul 2026 – Present*

Supervised by Prof. Nevin Lianwen ZHANG

- Designed and implemented a trust-weighted review summarisation module using aspect clustering, pros/cons extraction, and self-critique verification with LLMs.
- Built a deterministic data acquisition pipeline with schema validation and commercial API fallback, collecting and processing product/review data across platforms.
- Developed tool schemas and contributed to the agent orchestration layer for open-ended query decomposition and trust-aware routing.

Cathay Hackathon 2025 — Participant *Sep 2025 – Oct 2025*

- Proposed a RAG-based AI chatbot for airline transit information; converted UX challenges into technical proposals under time constraints.
- Applied design thinking and collaborative brainstorming to improve passenger services.

INTERNSHIP EXPERIENCE

Summer AI Intern *Hong Kong, China*
ESD Services Limited (ESDlife) *July 2026 – Aug 2026*

- Perform UAT Testing
- Implement 2 social media web scrapers (Facebook Instagram)
- Analyse social media trend topic using various NLP, LLM, and ML method

AI Programmer *Hong Kong, China*
Centre for Intelligent Multidimensional Data Analytics Limited *June 2025 – Aug 2025*

- Collaborate with senior engineers to work on some new web development projects (i.e. CV system for engineers)
- Working on backend development using PostgreSQL and writing several APIs for the React Native frontend using JavaScript, Node.js, etc

AWARDS AND RECOGNITIONS

Dean of Engineering Scholarship (HKUST) *Nov 2023*

- In recognition of excellent academic achievement in HKDSE.

Outstanding TKPSS Graduate Award (Tin Ka Ping Foundation) *Oct 2023*

- In recognition of remarkable academic achievement in HKDSE.

UNDERGRADUATE RESEARCH PROJECT EXPERIENCE

Heat Index Estimation — UROP, HKUST

Jun 2025 – Present

Supervised by Prof. FUNG, Jimmy Chi Hung; co-supervised by Dr. CHAN, Jimmy Wai Man, and Dr. Tilman Leo Hohenberger (starting from 2026 Summer)

2026 Fall

Sep 2026 – Present

- Estimate spatial temperature across the whole HK region.
- Use various neural network models for land cover classification.
- Plan and operate drones to capture DJI thermal JPG images in specified regions.

2026 Summer

Jun 2026 – Aug 2026

- Developed a Python script for automatic geo-referencing of thermal images (DJI thermal JPG → geo-coded raster).
- Operated drones to capture DJI thermal JPG images.

2026 Spring

Jan 2026 – May 2026

- Developed a 24-hour building shadow system across the HKUST campus and expanded the domain to the whole HK region.
- Optimized building shadow computation using multi-threading and multi-processing techniques.

2025 Fall

Sep 2025 – Dec 2025

- Developed an urban building shadow analysis framework with seamless map merging for dense urban environments.
- Investigated sunlight pattern determination and visualization at minute-level intervals with high temporal resolution.
- Investigated map merging strategies without boundary artifacts.
- Performed GIS work on building shadows of different morphologies, solar irradiance, and HK map extraction.

2025 Summer

Jun 2025 – Aug 2025

- Predicted localized temperature exposure across different times and locations using a Sky-View Factor approach.
- Performed GIS work on multi-source geospatial data (3D building morphology, building shadows) and microclimate parameters (solar radiation, temperature) using QGIS, CSDI portal, Open3Dhk, and Overpass Turbo.

COURSE HIGHLIGHT

- **AI & Data:** Artificial Intelligence (Python), Large Language Model, AI Agents, Image Processing, Machine Learning, NLP, Data Mining, Reinforcement Learning
- **Programming & Systems:** Object-Oriented Programming (C++), Software Engineering (Java), Operating Systems (Linux), Data Structure, Design and Analysis of Algorithms, Computer Organizations
- **Math & Statistics:** Statistical Computing (R), Nonparametric Statistics, Probability, Honor Discrete Mathematics
- **Web & Others:** Web Development (PostgreSQL, JavaScript, Node.js, React Native), GIS (ArcGIS, QGIS), Vibe Coding, Design Thinking

SOFT SKILLS

- **Willing to Learn:** Proactively acquired geospatial and environmental knowledge for a research project, despite having no prior background in the field.
- **Time Management:** Punctual, attends meetings early; consistently finishes tasks days before deadlines.
- **Organizational:** Proficient in organizing workloads and prioritizing tasks using planning tools to optimize productivity.
- **Proactive Self-learner:** Completed Coursera and Kaggle courses; self-taught the latest AI tools and concepts (e.g., LLMs, AI Agents), Virtual Machines, JavaScript, and Node.js outside of work hours.
- **Open to Criticism:** Embraces feedback as a growth opportunity; actively integrates constructive criticism to improve performance.

ENRICHMENT ACTIVITIES AND LEADERSHIP ROLES

Tin Ka Ping Secondary School F.5 Mathematics (Extended Part Module 2) Enhancement Class Tutor *Feb 2025 - May 2025*

- Prepare course outlines, handouts, and review and provide feedback on the coursework for 23 participating students who are taking M2 as one of their elective subjects in HKDSE
- Deliver a total of 7 tutorials for the high school students based on the course materials prepared

Tutor of the Revision Class for Special Education Needs (SEN) students *Apr 2024 - May 2024*

- Giving academic support and guidance to the SEN students

Tin Ka Ping Secondary School F.6 Chemistry Enhancement Class Tutor *Oct 2023 - Feb 2024*

- Prepare course outlines, handouts, and review and provide feedback on the coursework for 1-2 participating students who are taking Chemistry as one of their elective subjects in HKDSE
- Deliver a total of 10 tutorials for the high school students based on the course materials prepared